

Total No. of Questions : 8]

[Total No. of Printed Pages : 1

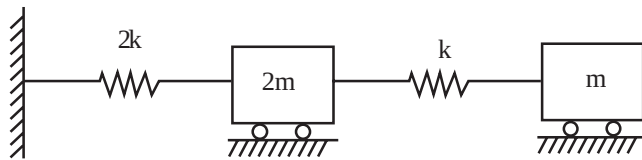
Roll No

CE-8041 (GS)
B.E. VIII Semester Examination, June 2020
Grading System (GS)
Structural Dynamics and Earthquake Engg.
(Elective-II)
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. What do you understand by Earthquake? Discuss various causes of Earthquakes.
2. What do you mean by Hysteretic damping and its importance in Earthquake Resistant design?
3. Write algorithm for step-by-step solution of a linear system using Newmark's method.
4. Determine the natural frequencies of the vibrating system shown in Fig.



Figure

5. Using neat sketches, explain the IS code requirements for ductile detailing of RC building.
6. What do you mean by under damped system? Explain.
7. Discuss the Eigen value problem and various methods for solving it.
8. Explain the following terms:(any two)
 - i) Fundamental natural period of vibration
 - ii) Effect of ground motion on structures
 - iii) Design seismic forces
